
AKTU
SEM-2
ENVIRONMENT & ECOLOGY
UNIT-2
NATURAL RESOURCES
One Shot
COMPLETE THEORY
IMPORTANT DIAGRAMS
EXAM POINTS
EASY TO REVISE
PDF
NOTES
ONE SHOT PDF NOTES
AKTU PYQs COVERED
PREVIOUS YEAR QUESTIONS
LUCKNOW

Environment and Ecology

AKTU Syllabus

Unit-2 : Natural Resources

Natural Resources: Introduction, Classification.

Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases; Fluoride and Arsenic Problems in Drinking Water.

Mineral Resources; Material Cycles; Carbon, Nitrogen and Sulfur cycles.

Energy Resources; Conventional and Non conventional Sources of Energy.

Forest Resources; Availability, Depletion of Forests, Environment impact of forest depletion on society.

Natural Resources (प्राकृतिक संसाधन)

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Natural resources are the materials or substances that occur in the natural environment which are used for various purposes.

Examples of *Natural resources* are-

Water, Coal, Oil, Sunlight, Sea/Ocean, Wind, Natural Gas, Minerals, Nuclear etc.

These resources can be classified into two main categories –

1. **Renewable Resources** (अक्षय संसाधन):
2. **Non-renewable Resources** (अनवीकरणीय संसाधन):

Major reasons for depletion of natural resources

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Reason	Description
Overpopulation	**Increased demand for resources due to a growing global population.**
Deforestation	Clearing forests for agriculture, urban development, leading to habitat loss.
Overexploitation	**Excessive use of resources for economic gain, surpassing natural replenishment rates.**
Pollution	Contamination of air, water, and soil, which degrades natural resources and ecosystems.
Climate Change	**Altered weather patterns affect resource availability and ecosystems' health.**
Industrialization	Rapid industrial growth increases resource consumption and waste generation.
Agricultural Practices	**Intensive farming and poor land management lead to soil degradation and loss of fertility.**
Urbanization	Expansion of cities reduces available land and increases resource consumption.
Mining Activities	**Extraction of minerals depletes non-renewable resources and damages the environment.**
Water Mismanagement	Overuse and pollution of water resources lead to scarcity and degradation.

Sustainability of Natural Resources

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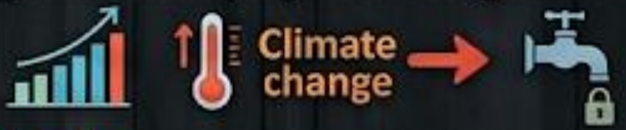
Strategy	Actions
Conservation and Efficient Use	**Reduce consumption, Recycle and reuse materials, Practice sustainable agriculture**
Renewable Energy	Use renewable energy sources, Improve energy efficiency
Protecting Ecosystems	**Preserve biodiversity, Implement sustainable land use**
Policy and Legislation	Enforce environmental regulations, Provide incentives for sustainable practices
Education and Awareness	**Educate the public on sustainability, Engage communities in conservation efforts**
Technological Innovation	Develop green technologies, Build sustainable infrastructure
International Cooperation	**Participate in global environmental agreements, Support sustainable development in developing nations**

Availability of water on the earth

- Earth is often referred to as the "Blue Planet" due to its abundant water, but the distribution and availability of freshwater are uneven.
- About **97%** of the Earth's water is in the oceans (saltwater), leaving only about **3%** as freshwater.
- However, the majority of freshwater is locked in ice caps, glaciers, and underground, making only a small percentage readily accessible for human use.
- This limited availability, coupled with population growth and climate change, poses challenges for water security in various regions.



Availability of Water in India

- India is the wettest country in the world, but rainfall is highly uneven with space and time.
- Rainfall is high in the North-East but extremely low in Rajasthan.

Aspects (characteristics) of good quality water

- ✓ Maintaining good water quality is crucial for sustaining life, safeguarding ecosystems, and supporting various human activities.
- ✓ Monitoring and managing water quality are essential for preserving resources.

These are the following characteristics of good-quality water:

1. It is transparent, colorless and odorless.
2. It has sufficient oxygen concentration for marine life to survive.
3. It is free from bacteriological contamination.
4. It is free from any water pollution.
5. It is free from excessive nutrients like N, P, etc.
6. It is pure and fit for the intended use.

1.


2.


3.


4.


5.


6.


Use and effects of **Over Utilization of surface and ground water sources**

AKTU 2022-23 Even

→ The overutilization of surface and groundwater sources can have significant impacts on the environment, society, and the economy.

Overutilization of Surface Water Sources Uses:

- **Agriculture:** Surface water is extensively used for irrigation. 
- **Domestic Use:** Drinking water, sanitation, and household needs. 
- **Industrial Use:** Factories and power plants use surface water for cooling and processing. 
- **Recreation:** Lakes and rivers are used for boating, fishing, and tourism. 
- **Hydropower:** Generation of electricity through dams and water turbines. 

Effects:

✓ **Reduced Water Availability:** Overdrawing from rivers and lakes can lead to lower water levels, affecting ecosystems and human needs.

...Use and effects of Over Utilization of surface and ground water sources

- ✓ **Land Subsidence (भू-धंसाव):** Removal of large amounts of groundwater can cause the ground to sink, damaging infrastructure.

- ✓ **Saltwater Intrusion:** In coastal areas, over-pumping can lead to saltwater contaminating freshwater aquifers.

- ✓ **Reduced Surface Water Flow:** Groundwater and surface water are interconnected; lowering groundwater levels can reduce the flow of rivers and streams.

- ✓ **Quality Degradation:** Overuse can cause contaminants to enter the aquifer, reducing water quality.

Sustainable Practices

- → Efficient Water Use
- → Rainwater Harvesting
- → Regulation and Management
- → Public Awareness

Water Borne Diseases (जल जनित रोग)

AKTU 2013-14, 2017-18, 2023-24

Waterborne diseases are spread or caused by the consumption of **contaminated water** (contaminated with pathogenic microorganisms) and **preparation of food by contaminated water**. Waterborne diseases include - Cholera, Typhoid Fever, Dysentery, Giardiasis etc.

1.Cholera (हैजा)

- ✓ **Cause:** Vibrio cholerae bacteria
- ✓ **Transmission:** Consumption of contaminated water or food.
- ✓ **Effects:** Watery diarrhea, vomiting, and leg cramps (पैर में ऐंठन). In these people, rapid loss of body fluids leads to dehydration, shock and Kidney failure etc.

- ✓ **Preventive Measures:** Water purification, sanitation, and hygiene practices, Eating fresh healthy foods.

Water Borne Diseases (जल जनित रोग)

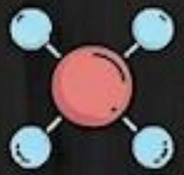
2. Typhoid Fever (टाइफाइड बुखार)

AKTU 2013-14, 2017-18, 2023-24

- ✓ **Cause:** Salmonella typhi bacteria.
- ✓ **Transmission:** Consumption of contaminated water or food.
- ✓ **Effects:** High fever accompanied with diarrhea or vomiting.
- ✓ **Preventive Measures:** Safe drinking water, improved sanitation, and vaccination, Eating fresh healthy foods.

3. Dysentery (पेचिश)

- ✓ **Cause:** Various bacteria, parasites, or viruses.
- ✓ **Transmission:** Contaminated water or food.
- ✓ **Effects:** Diarrhea, fever, nausea, vomiting, weight loss and stomach cramps (पेट में ऐंठन).
- ✓ **Preventive Measures:** Clean water supply, sanitation, and personal hygiene, Eating fresh healthy foods.

Water-Induced diseases (जल-प्रेरित रोग)

Water induced diseases are generally caused by protozoa.

These are the following common water-induced diseases:

1. Malaria

Cause: Plasmodium parasites transmitted by mosquitoes breeding in stagnant water.

Effects: Vomiting, and diarrhea. If not promptly treated, the infection can become severe and may cause kidney failure, seizures, mental confusion, coma, and death.

Preventive Measures: Mosquito control, bed nets, and antimalarial drugs.

2. Dengue

Cause: Mosquito Biting.

Effects: Nausea, vomiting, Aches and pains (eye pain, typically behind the eyes, muscle, joint, or bone pain) etc.

Preventive Measures: Mosquito control, bed nets, and ant parasitic drugs.

Causes and Prevention measures of water-related Disease

Causes of water related disease

1. Contaminated Water Sources
2. Inadequate Water Treatment
3. Lack of Sanitation
4. Poor Personal Hygiene
5. Stagnant Water
6. Lack of Access to Clean Water

Prevention measures of water-related disease

1. Water Treatment
2. Improved Sanitation
3. Hygiene Education
4. Vaccination
5. Access to Clean Water
6. Monitoring and Surveillance (Water Quality)
7. Community Awareness

Fluoride Problem in India

AKTU-2023-24, 2017-18

- ✓ The fluoride problem in India is a significant public health concern, primarily associated with high levels of naturally occurring fluoride in drinking water.
- ✓ While fluoride is essential for dental health in appropriate amounts, excessive intake, especially through drinking water, can lead to health issues. Many parts of India, particularly in the states of Rajasthan, Andhra Pradesh, Telangana, Gujarat, and Tamil Nadu, experience elevated fluoride concentrations in groundwater.

Causes

- ✓ **Geological Factors (भूगर्भीय कारक):** The presence of fluoride in the Earth's crust can result in leaching into groundwater.

- ✓ **Agricultural Practices:** The use of phosphate fertilizers and certain pesticides can contribute to elevated fluoride levels.

Effects on human health - Fluoride Problem in India

1. Dental Fluorosis

***Mottling of Teeth* (दांतों का धब्बेदार होना):** Prolonged exposure to high fluoride levels during tooth development can lead to dental fluorosis, characterized by discoloration, pitting, and mottling of tooth enamel.

2. Skeletal Fluorosis

***Bone and Joint Issues* (हड्डी और जोड़ की समस्याएँ):** Long-term exposure to high fluoride concentrations can cause skeletal fluorosis, a condition marked by pain and damage to bones and joints. It can result in stiffness, limited joint mobility, and, in severe cases, deformities.

Other Health Impacts:

Reduced Fertility: Some studies have explored potential associations between high fluoride exposure and reduced fertility, but findings are not consistent.

Preventive measures for Fluoride problems

1. **Water Quality Testing** - Regular testing of water sources for fluoride levels to identify areas with high concentrations.
2. **Alternative Water Sources** - Providing alternative sources of drinking water in fluoride-affected areas.
3. **Water Treatment** - Implementing water treatment methods, such as defluoridation techniques, to reduce fluoride levels in water.
4. **Public Awareness** - Raising awareness about the health risks associated with high fluoride intake and promoting safe water consumption practices.
5. **Dental Health Education** - Educating communities on dental hygiene practices to mitigate the risk of dental fluorosis.
6. **Government Policies** - Implementing and enforcing policies to regulate and monitor fluoride levels in drinking water.

Arsenic problem in India

- The arsenic problem in India is a significant environmental and public health issue, particularly in certain regions where groundwater contains elevated levels of arsenic.
- Arsenic contamination in drinking water poses serious health risks to the population.
- The problem is prevalent in several states, including West Bengal, Bihar, Uttar Pradesh, Jharkhand, and Assam.

Causes

- ✓ **Geological Factors (भूगर्भीय कारक):** Arsenic occurs naturally in tn release arsenic into groundwater.
- ✓ **Anthropogenic Activities (मानवजनित गतिविधियाँ):** Human activities such as mining, industrial pesticides can contribute to arsenic cestins.

Effects on human health -Arsenic problem in drinking water

1. Skin Lesions (त्वचा के घाव): Prolonged exposure to arsenic-contaminated water can lead to skin lesions, including hyperpigmentation , keratosis, and the development of hard patches on the skin.

2. Cancer Risk: Chronic exposure to arsenic is associated with an increased risk of various cancers, including skin, lung, bladder, and kidney cancers.

3. Cardiovascular Effects (हृदय संबंधी प्रभाव): Arsenic exposure has been linked to cardiovascular diseases, including hypertension and atherosclerosis.

4. Neurological Impacts (तंत्रिका तंत्र पर प्रभाव): Some studies suggest potential neurological effects, such as cognitive deficits and developmental issues in children exposed to high levels of arsenic.

5. Respiratory Problems (श्वसन समस्याएँ): Long-term exposure to arsenic may contribute to respiratory issues, including chronic bronchitis and respiratory tract infections.

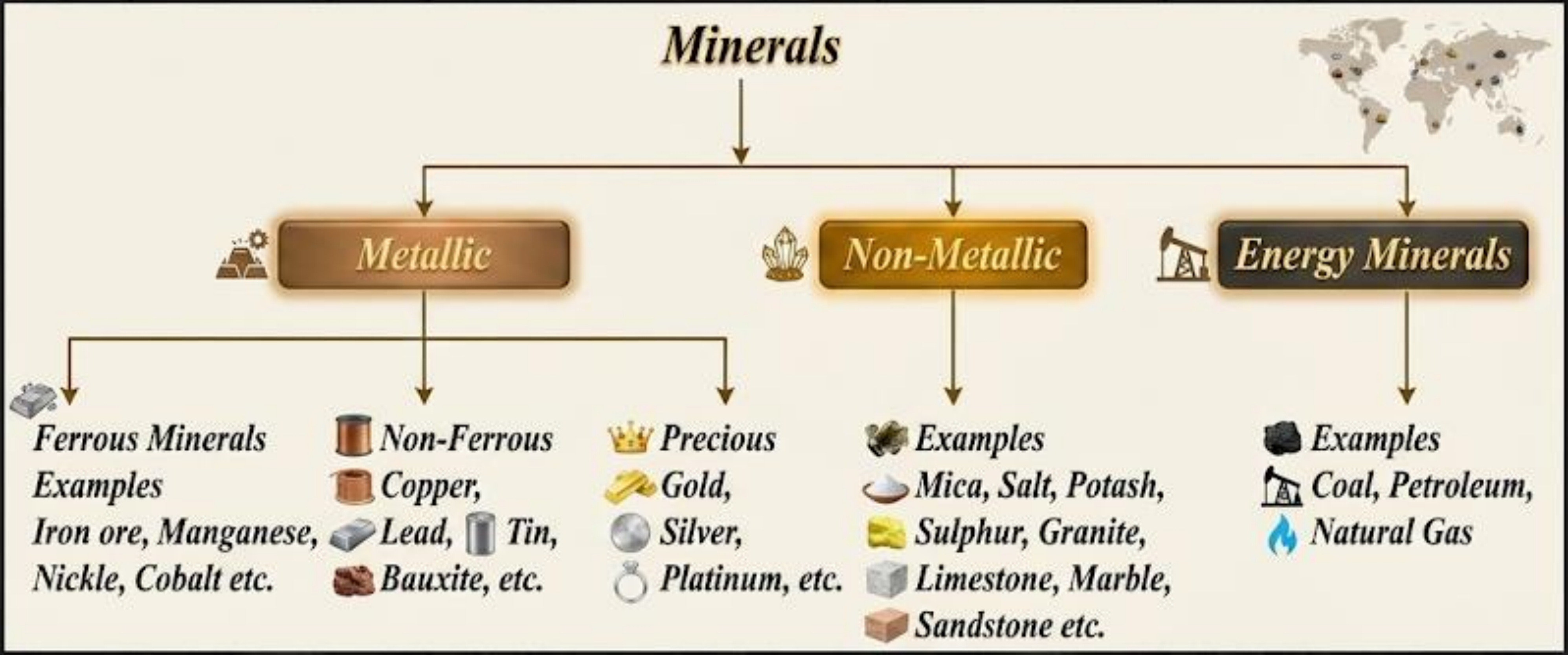
Preventive measures for Arsenic Problems

1. **Water Quality Testing** - Regular testing and monitoring of groundwater to identify arsenic - contaminated areas.
2. **Alternative Water Sources** - Providing alternative sources of safe drinking water in arsenic-affected regions.
3. **Public Awareness** - Raising awareness about the health risks associated with arsenic exposure and promoting safe water consumption practices.
4. **Government Policies & Regulations** - Development and enforcement of policies to regulate arsenic levels in drinking water and industrial discharges.
5. **Healthcare Interventions** - Offering medical screenings and healthcare services in affected areas to identify and address health issues related to arsenic exposure.
6. **Community Engagement** - Involving communities in decision-making processes, encouraging community-led initiatives, and fostering local participation in addressing the arsenic problem.

Mineral Resources

→ Mineral resources are natural substances found in the Earth's crust that have economic value.

→ **Examples** of mineral resources include iron ore, copper, gold, coal, oil, natural gas, limestone, and many others.



Effects of Mineral Extraction on the Environment

→ While mineral extraction is essential for economic development and various industrial processes, it has significant impacts on the environment.

→ The extraction and processing of minerals can lead to **environmental challenges**, affecting ecosystems, water quality, air quality, and biodiversity.

These are the following effects of mineral extraction on the environment –

1. Habitat Destruction (आवास का विनाश):

→ Mining activities can result in the destruction of natural habitats and ecosystems.

→ Spills and leakages from processing facilities can release toxic chemicals, impacting aquatic life.

4. Air Pollution (वायु प्रदूषाण):

→ Dust particles and heavy maching of ores can release harmful gases and particulate matter.

5. Land Degradation (भूमि क्षरण):

→ Loss of arable land and topsoil due to mining.

Effects of Mineral Extraction on the Environment

3. Water Pollution:

→ Runoff from mining sites may contain pollutants, such as heavy metals, sediments, and chemicals, which can contaminate nearby rivers and streams.

→ This pollution can harm aquatic ecosystems and effect the quality of drinking water.

4. Air Pollution:

→ Dust and particulate matter generated during mining and mineral processing activities can contribute to air pollution.

→ Emissions from equipment and transportation also lead air pollution.

5.Deforestation:

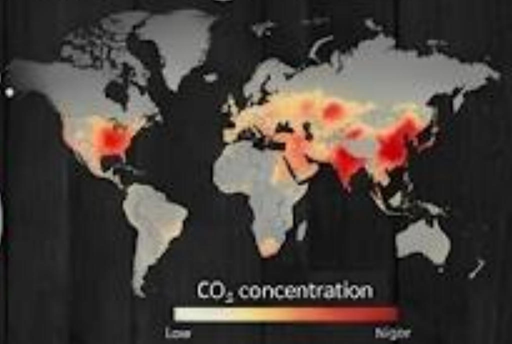
→ In some cases, mining operations may require the clearing of forests.

→ Deforestation can result in the loss of biodiversity, disruption of ecosystems, and contribute to climate change.

Effects of Mineral Extraction on the Environment

6. Greenhouse Gas Emissions:

→ The extraction and processing of certain minerals, such as coal and oil, can release significant amounts of greenhouse gases, contributing to global warming and climate change.



7. Resource Depletion:

→ Unsustainable mining practices can lead to the depletion of finite mineral resources, impacting future generations and necessitating the development of alternative, sustainable sources.

8. Social and Cultural Impacts:

→ Mining activities can also have social and cultural impacts, including the displacement of communities, changes in traditional land use, and conflicts over resource access.

Material Cycle/Biogeochemical Cycle

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→ Material cycles are the movement and transformation of elements and compounds through different environmental compartments, including the atmosphere, land, water bodies, and living organisms.

→ Three important material cycles are:

1. Carbon cycle

2. Nitrogen cycle

3. Sulfur cycle

→ The carbon cycle, nitrogen cycle, and sulfur cycle are vital for ecological balance and supporting biological processes as they regulate the distribution and availability of essential elements, control climate by managing greenhouse gas concentrations, and guide sustainable resource management.

Biogeochemical cycle

→ It is a cycle in which essential elements and compounds are exchanged among within the earth's atmosphere, hydrosphere, lithosphere, and biosphere.

→ These cycles ensure the movement of nutrients like carbon, nitrogen, oxygen, phosphorus, and sulfur through the living and non-living components of the ecosystem, facilitating the sustainability of life on Earth.

Carbon Cycle

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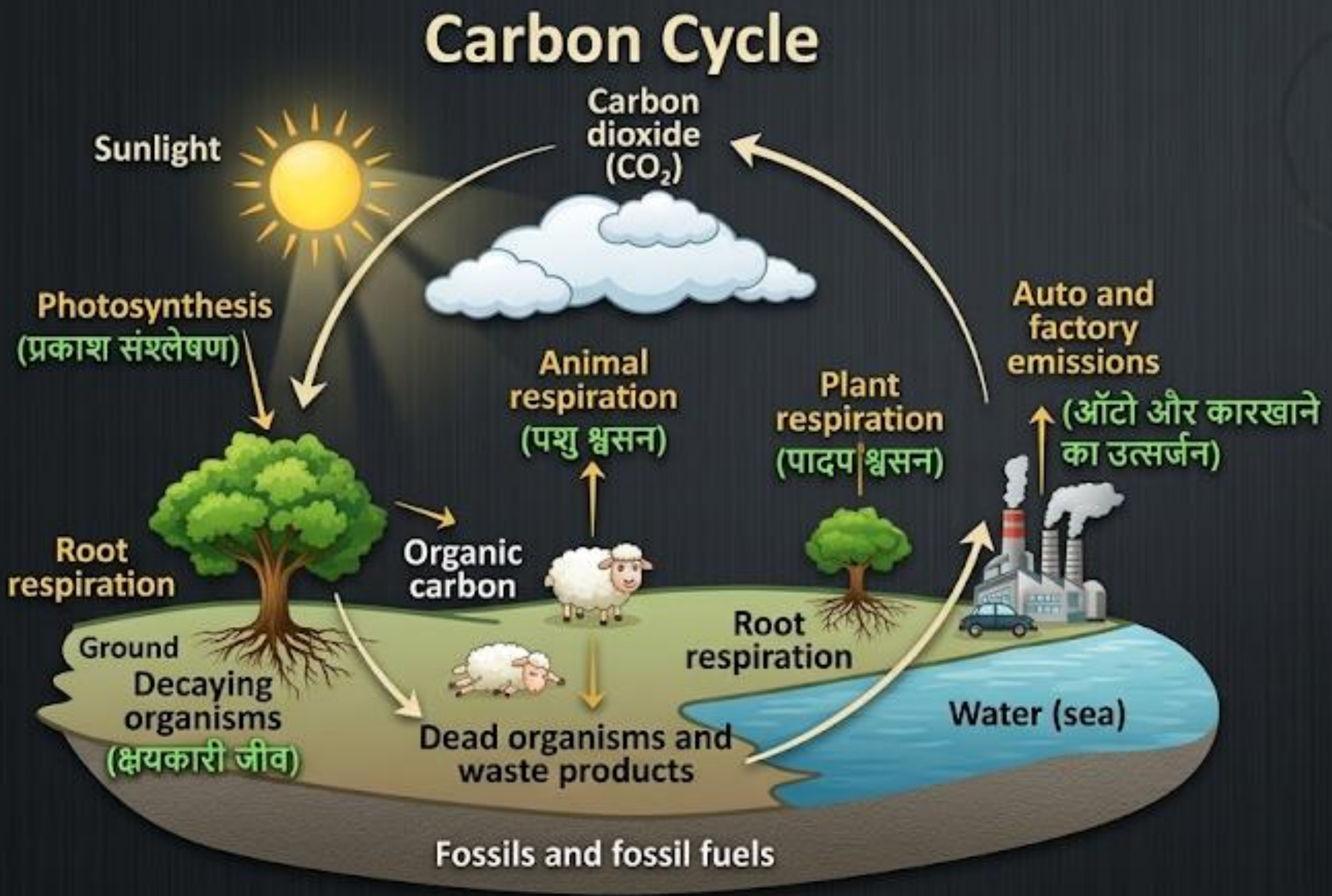


Carbon Cycle

AKTU 2014-15, 2017-18, 2022-23

The carbon cycle is a crucial process in the Earth's ecosystem that describes how carbon atoms move through various components of the Earth system.

Photosynthesis:

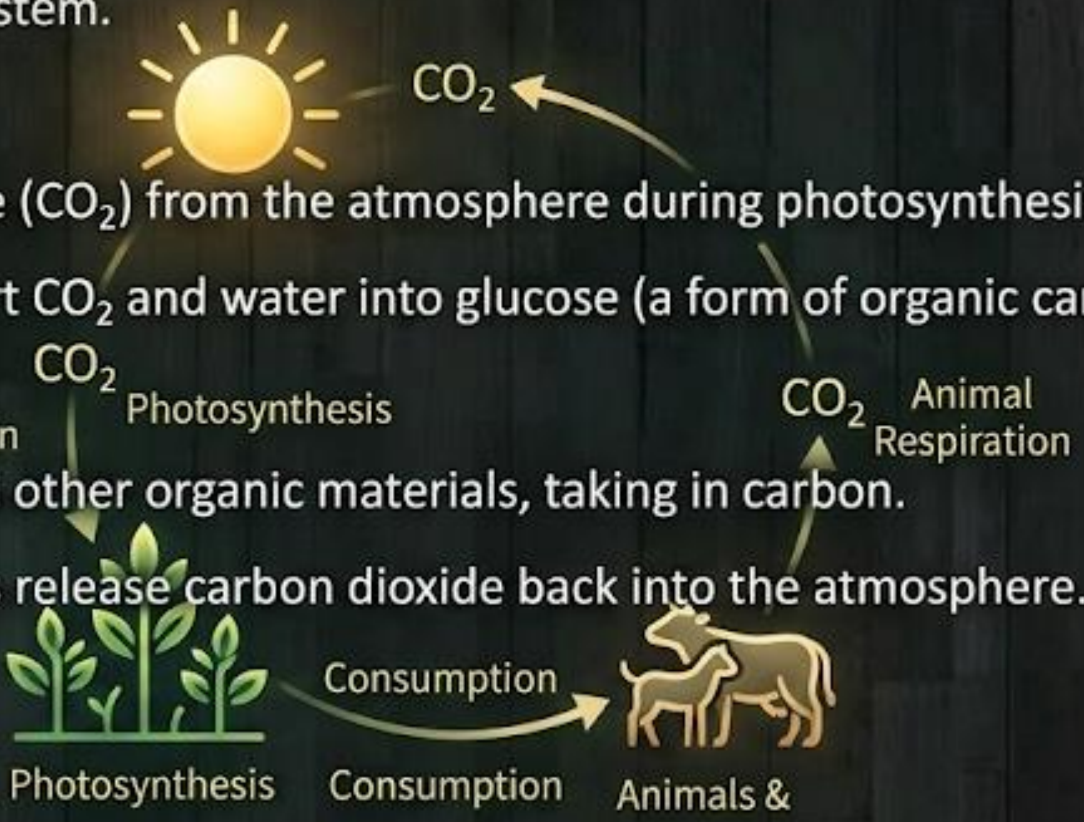
→ Plants absorb carbon dioxide (CO_2) from the atmosphere during photosynthesis.

→ Using sunlight, plants convert CO_2 and water into glucose (a form of organic carbon) and oxygen.

Animal Respiration:

→ Animals consume plants and other organic materials, taking in carbon.

→ Through respiration, animals release carbon dioxide back into the atmosphere.



Plant Respiration:

→ Plants also respire, releasing some of the carbon dioxide they have absorbed back into the atmosphere

Carbon Cycle

Decaying Organisms:

→ When plants and animals die, decomposers like bacteria and fungi break down their organic material.

→ This process releases carbon back into the soil and atmosphere.

Dead Organisms and Waste Products:

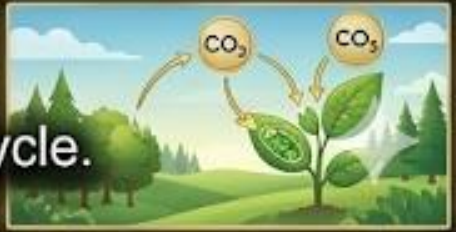
→ Dead plants and animals, along with their waste products, contribute organic carbon to the soil.

Fossils and Fossil Fuels:

→ Over millions of years, some of the organic carbon from dead organisms becomes buried and forms fossil fuels (like coal, oil, and natural gas).

Auto and Factory Emissions:

→ Burning fossil fuels for energy releases stored carbon back into the atmosphere as carbon dioxide.



Carbon Dioxide in the Atmosphere:

→ CO_2 in the atmosphere is absorbed by plants for photosynthesis, continuing the cycle.

Nitrogen Cycle

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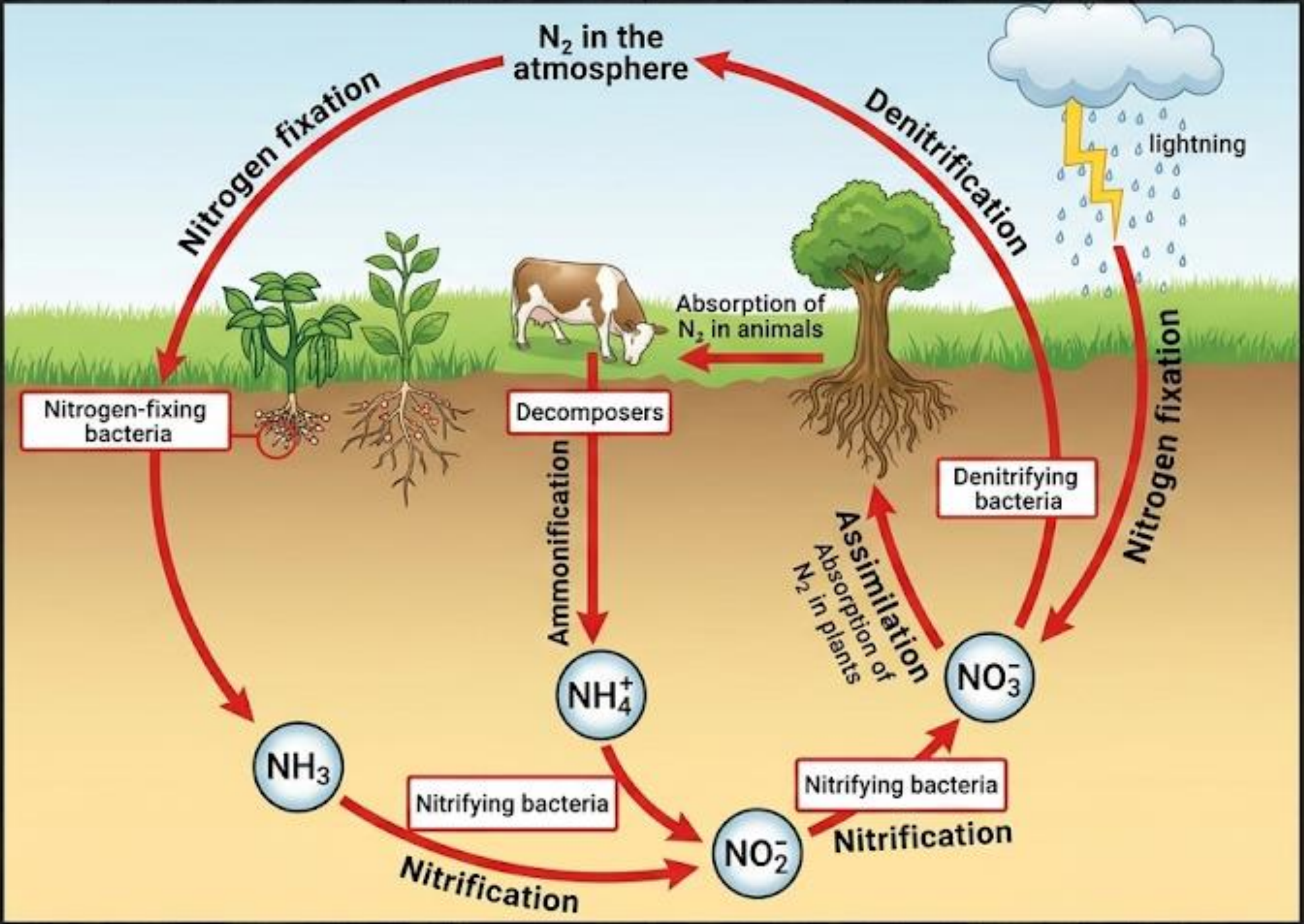

Nitrogen Cycle

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→ The nitrogen cycle is a crucial ecological process that describes the movement of nitrogen through the atmosphere, soil, plants, animals, and microorganisms.

Nitrogen Fixation:

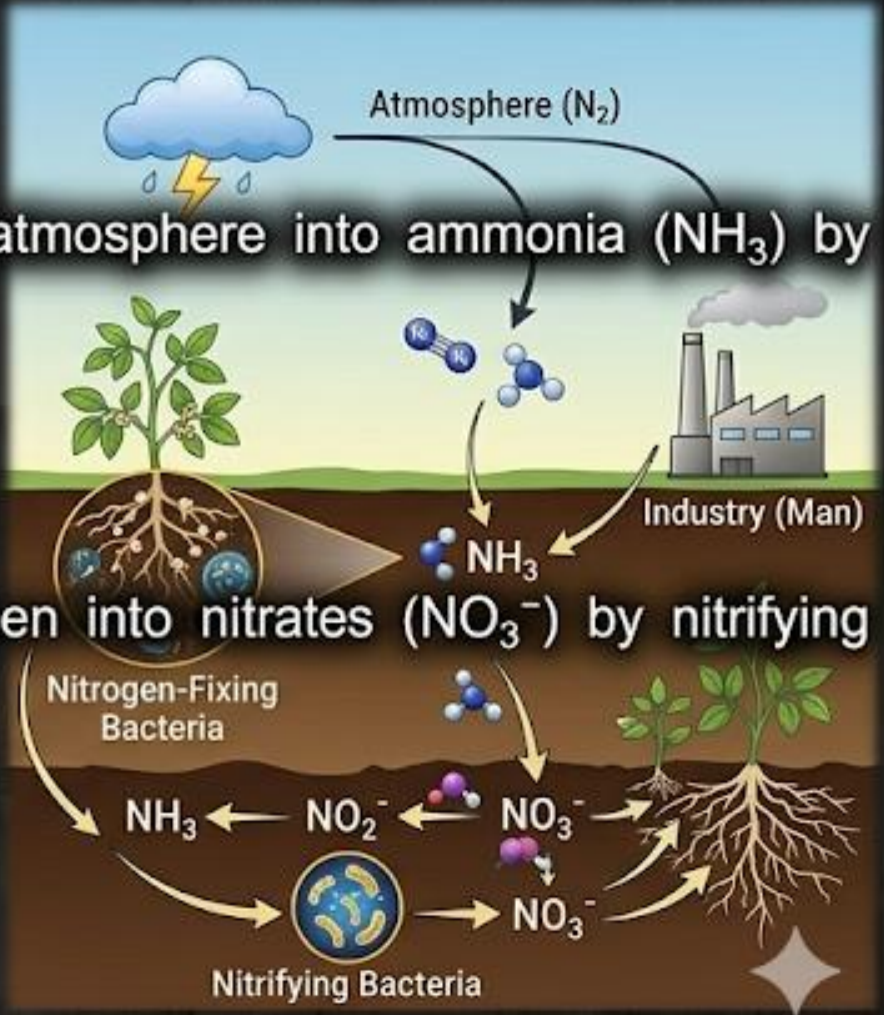
→ It is the process of converging nitrogen gas (N_2) from the atmosphere into ammonia (NH_3) by nitrogen-fixing bacteria in the soil or root of plants.

→ This process can also occur through man(industry) or lightning.

Nitrification:

→ Ammonia in the soil is converted into nitrites (NO_2^-) and then into nitrates (NO_3^-) by nitrifying bacteria. $\rightarrow NO_2^- \rightarrow NO_3^-$

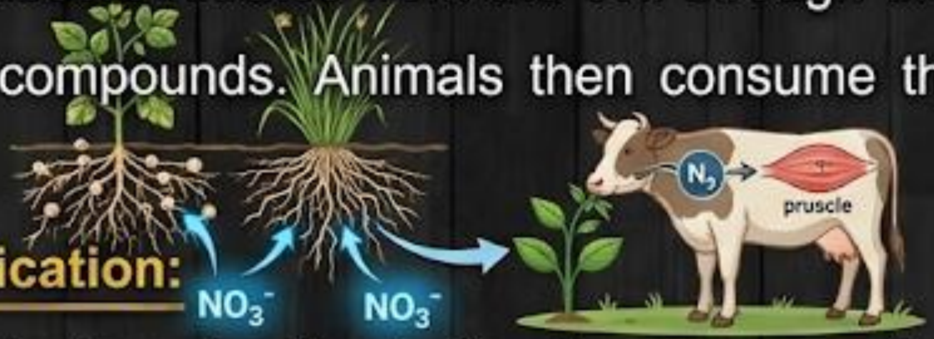
→ These forms are more accessible to plants.



...Nitrogen Cycle

Assimilation:

→ Plants absorb nitrates from the soil through their roots and use them to build proteins and other essential compounds. Animals then consume these plants, incorporating nitrogen into their own bodies.

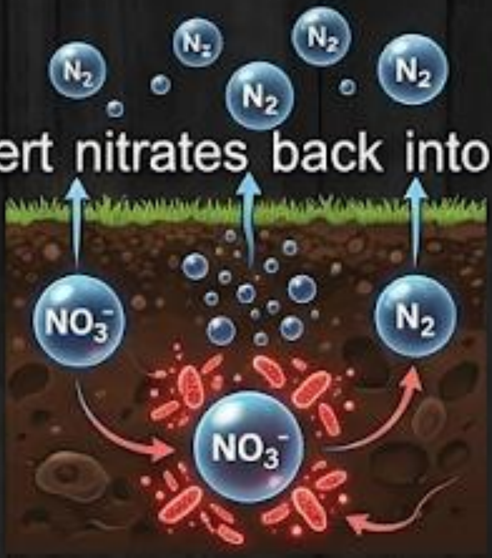


Ammonification:

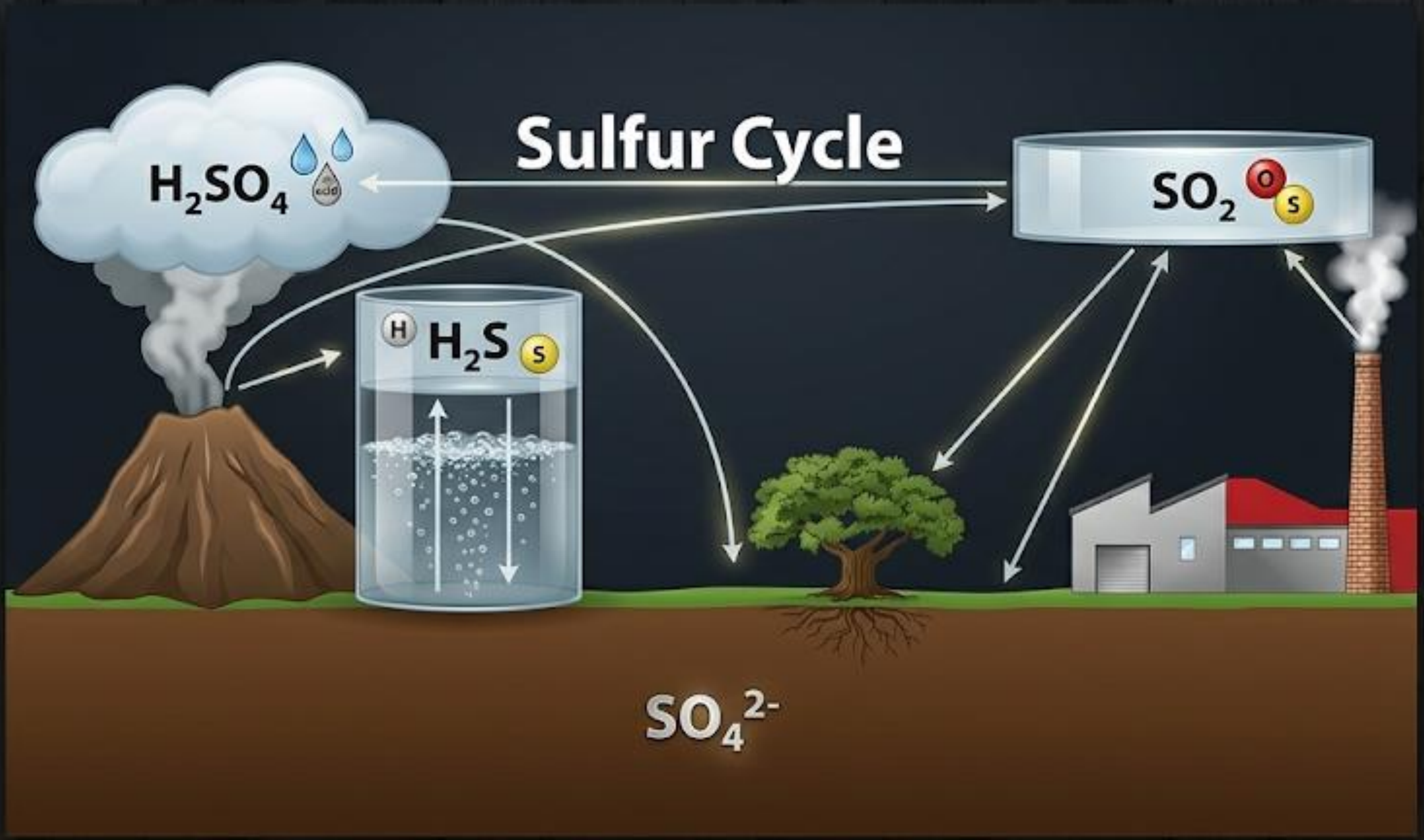
→ When plants and animals die, decomposers (bacteria and fungi) break down their organic matter, converting nitrogen back into ammonia or ammonium ions (NH_4^+), which return to the soil.

Denitrification:

→ Denitrifying bacteria in the soil convert nitrates back into nitrogen gas (N_2), releasing it into the atmosphere and completing the cycle.



Sulfur Cycle



Sulfur Cycle

→ The sulfur cycle is a biogeochemical cycle that describes the movement of sulfur through the lithosphere, hydrosphere, biosphere, and atmosphere. (with proper subscript SO_2)

Atmospheric Sulfur:

→ Sulfur dioxide (SO_2) is released into the atmosphere from volcanic eruptions, burning fossil fuels, and other industrial processes.

→ Sulfur gases can also be produced by the decay of organic matter.

Acid Rain:

→ Sulfur dioxide (SO_2) and other sulfur compounds in the atmosphere can combine with water vapor to form sulfuric acid (H_2SO_4), leading to acid rain.

→ Acid rain returns sulfur to the Earth's surface, where it can enter soils and water bodies.

...Sulfur Cycle

Sulfur in Soil and Plants: (मृदा और पौधों में सल्फर)

→ Plants absorb sulfate ions (SO_4^{2-}) from the soil, incorporating sulfur into essential amino acids and proteins.

→ Animals obtain sulfur by consuming plants.

Decomposition (अपघटन):

→ When plants and animals die, decomposers like bacteria and fungi break down their organic matter, releasing sulfur back into the soil as sulfate ions (SO_4^{2-}).

Sedimentation (अवसादन):

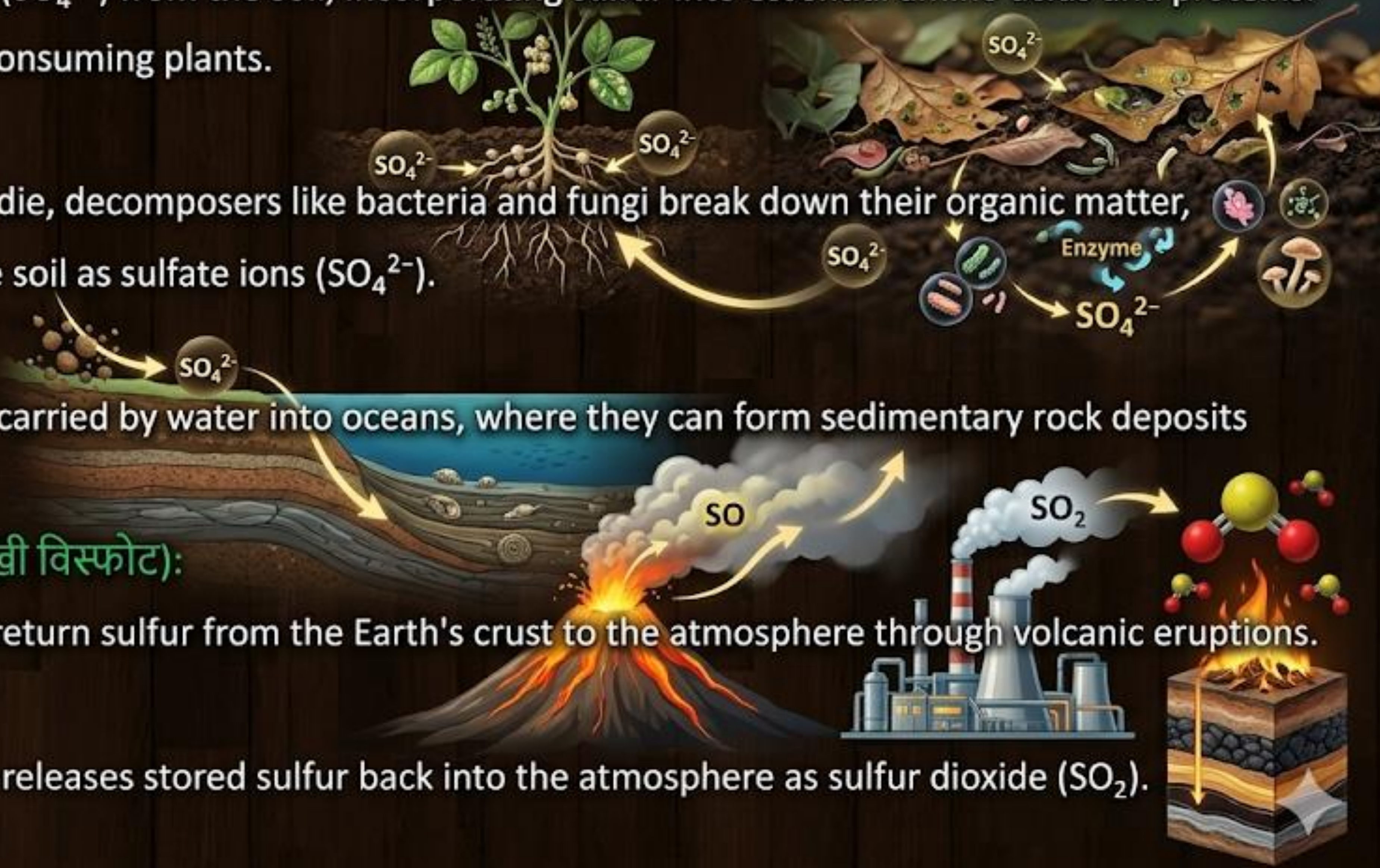
→ Sulfates in the soil can be carried by water into oceans, where they can form sedimentary rock deposits over time.

Volcanic Eruptions (ज्वालामुखी विस्फोट):

→ Geological processes can return sulfur from the Earth's crust to the atmosphere through volcanic eruptions.

Fossil Fuels (जीवाश्म ईधन):

→ The burning of fossil fuels releases stored sulfur back into the atmosphere as sulfur dioxide (SO_2).



Differences between Nutrient Cycling and Energy Flow

AKTU 2023-24 Odd

→ **Nutrient Cycling** involves the recycling of chemical elements (like carbon, nitrogen, phosphorus) within an ecosystem.

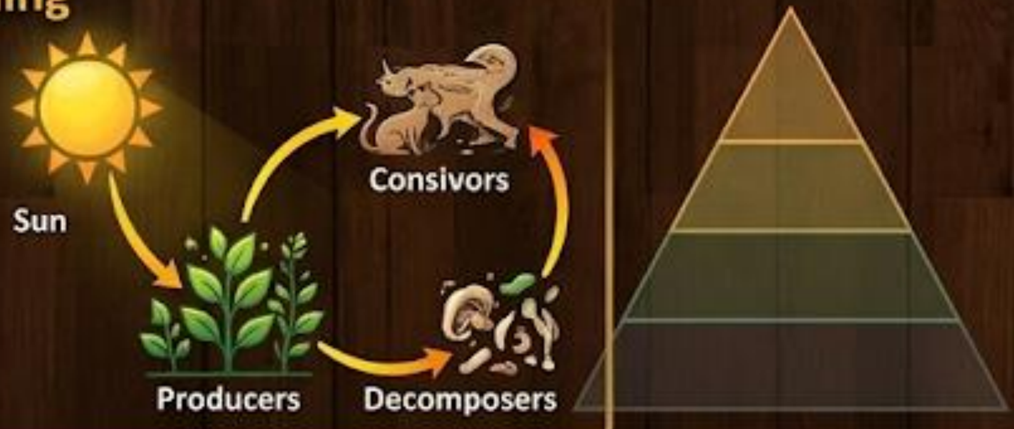
Example: The nitrogen cycle, where nitrogen moves through various forms (N_2, NH_3, NO_3^-) and between different ecosystem components. (N_2, NH_3, NO_3^-)

→ **Energy Flow** is the movement of energy through an ecosystem from the sun to producers and then to consumers and decomposers.

Example: Photosynthesis captures solar energy, which is then transferred consume plants, and so on up the food chain.

Importance of Nutrient Cycling

→ Sustaining Life
→ Soil Fertility
→ Ecosystem Stability
→ Productivity
→ Pollution Control
→ Climate Regulation



Importance of Nutrient Cycling

- Sustaining Life
- Soil Fertility
- Ecosystem Stability
- Productivity
- Pollution Control
- Climate Regulation

Non-renewable or conventional energy resources

AKTU 2013-14, 2022-23

Conventional Energy Resources: These are traditional sources energy that have been widely used for a long time.

→ Finite and Exhaustible:

→ Significant Environmental Impact:

→ They have significant environmental impacts, including **greenhouse gas emissions and pollution.**

→ Long Formation Time: These resources take **millions of years** within a human lifespan.

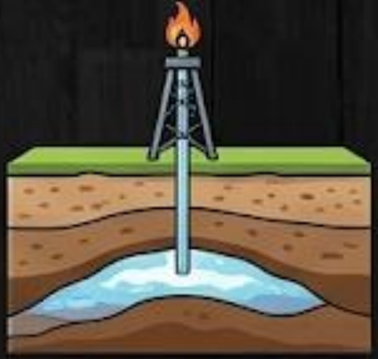
→ Slow Replenishment: These–span

These are ***non-renewable*** resources of energy.

Examples: fossil fuels sucal, oil, and natural gas, nuclear energy etc

Coal

Oil



Natural Gas

Nuclear Energy

Renewable or Non-conventional energy resources

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→ **Non-conventional energy resources** are renewable and sustainable sources of energy that can provide an alternative to traditional fossil fuels.

Examples:

→ Solar energy

→ Wind energy

→ Hydropower Energy

→ Biomass energy

→ Geothermal energy.

Non - Conventional energy resources - Solar energy

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Solar energy

→ *Solar energy* is a renewable energy resource that is obtained from the sun's radiation.

→ It is a clean and abundant source of energy that can be harnessed and converted into usable forms such as electricity or heat.

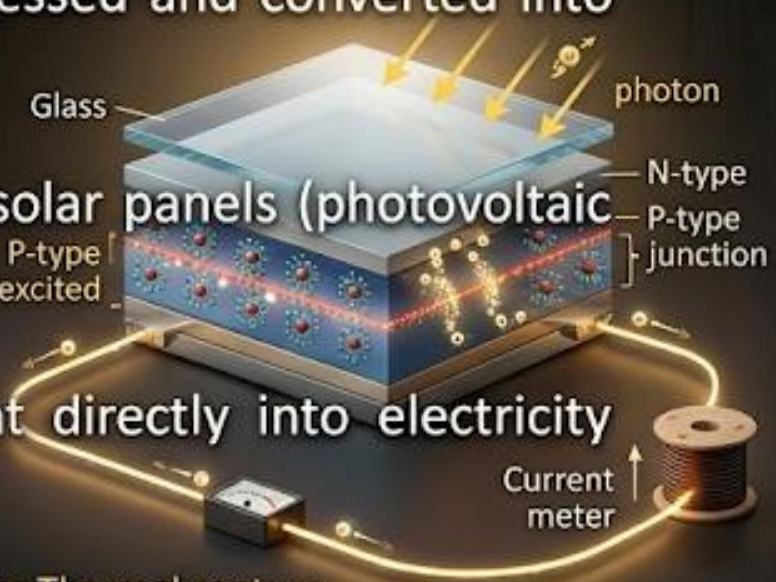
→ Solar energy is captured using various technologies, including solar panels (photovoltaic cells) and solar thermal systems.

Photovoltaic Cell: A solar cell is a device that converts sunlight directly into electricity through the photovoltaic effect.

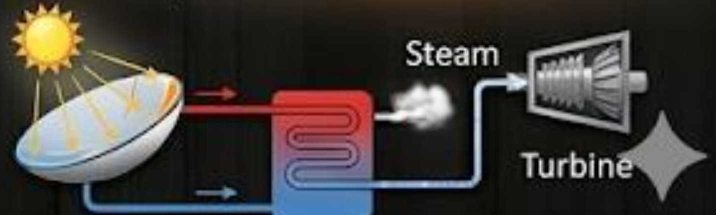
Photovoltaic (PV) Panel

Solar Thermal System

THE PHOTOVOLTAIC EFFECT IN ACTION



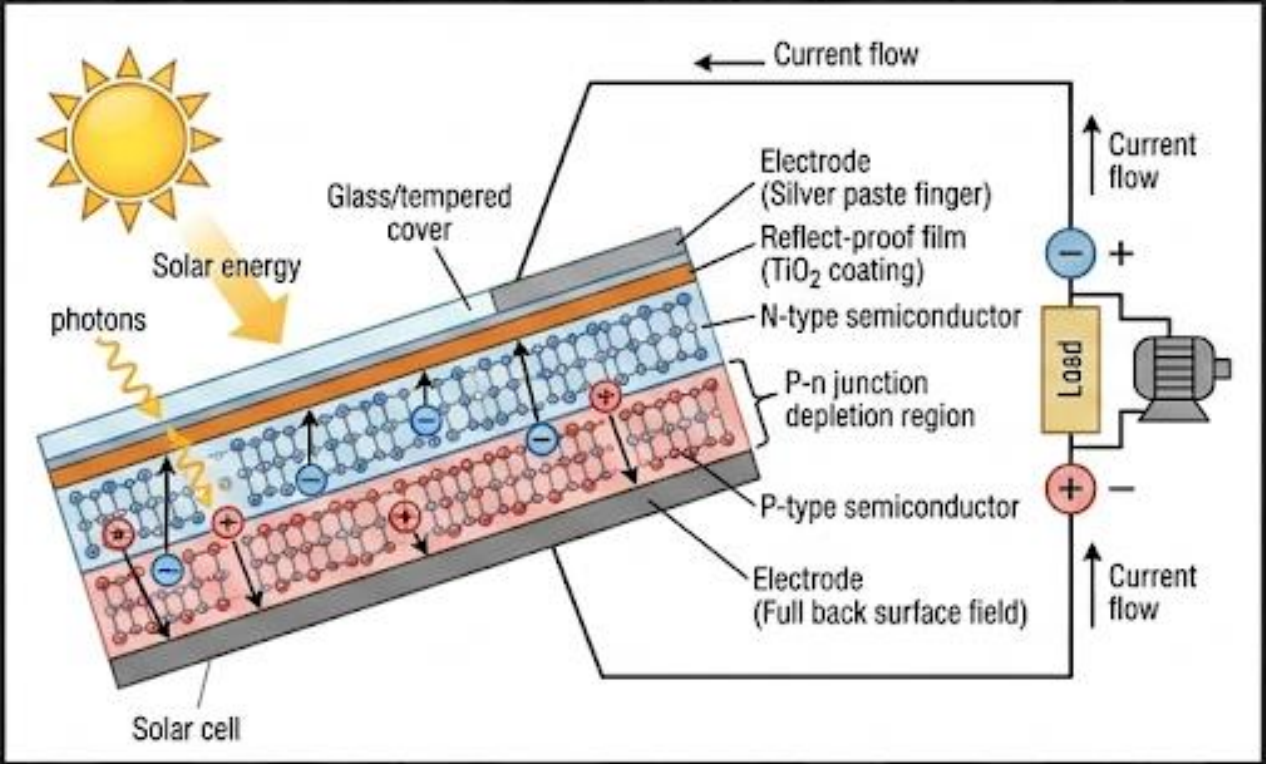
Solar Thermal capture



Photovoltaic Cell: A solar cell is a device that converts sunlight and ꝏ the alirous electricity throrough the photovoltaic effect.

...Non - conventional energy resources - Solar energy

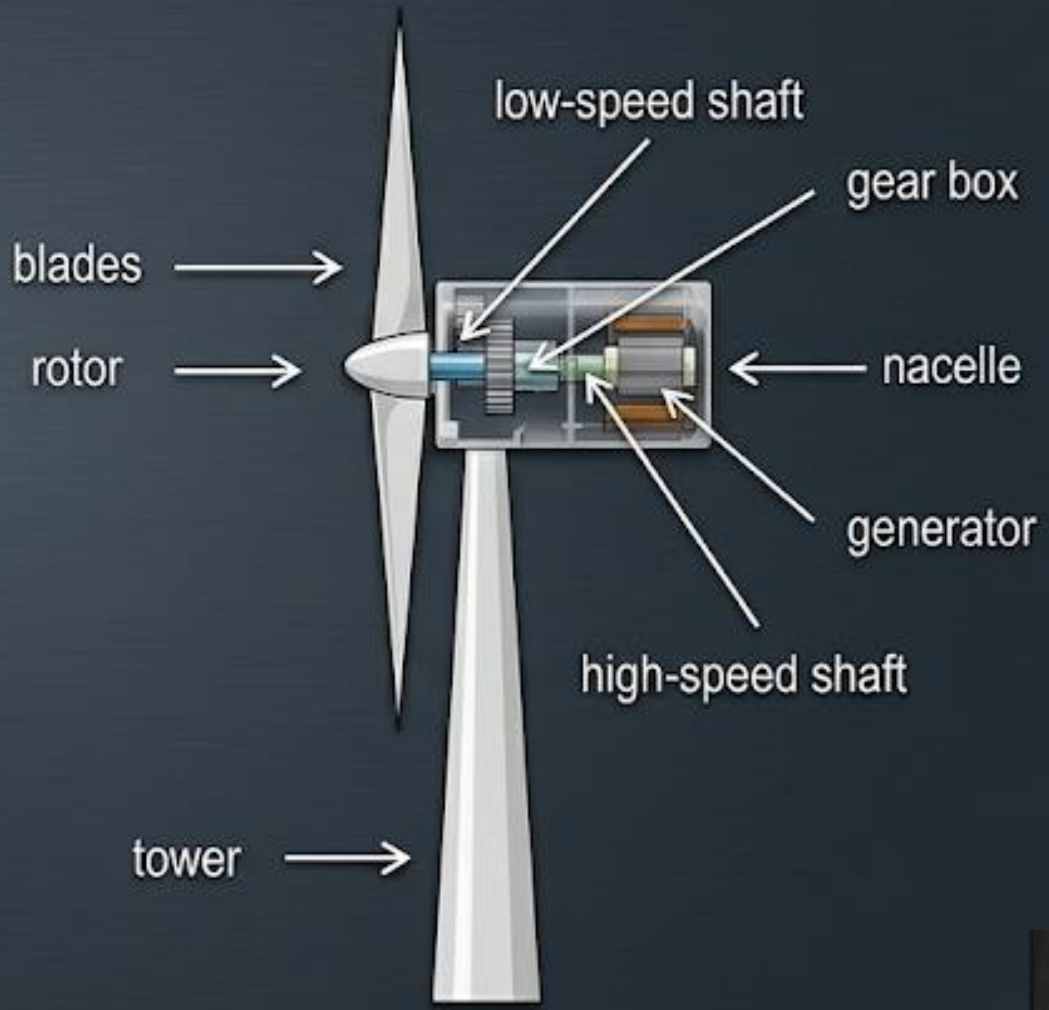
- The photovoltaic effect is the phenomenon where certain materials (semiconductor) generate an electric current when exposed to light.
- When sunlight hits the surface of the cell, the semiconductor material absorbs photons, causing the release of electrons and the creation of electron-hole pairs.
- Due to the internal electric field created by the p-n junction, an electric current is generated.



Non - Conventional energy resources - Wind energy

→ Wind energy is a form of renewable energy which is obtained from the natural flow of air in the Earth's atmosphere.

$$KE \xrightarrow[\text{Wind Turbine}]{} ME \xrightarrow{\text{Generator}} EE$$



...Non - Conventional energy resources - Wind energy

Wind Turbines:

→ Wind energy is captured using wind turbines, which are large structures with blades that rotate when the wind blows. Groups of turbines are used in wind power plants.

Kinetic to Mechanical Energy:

→ As the wind blows, it causes the blades of the turbine to rotate. This rotation of blades converts the wind's kinetic energy (energy of motion) into mechanical energy.

Mechanical to Electrical Energy:

→ Inside the turbine, the rotating blades turn a shaft connected to a generator. The generator converts the mechanical energy into electrical energy.

Electricity Distribution: → The electrical energy produced is then transmitted through power lines and distributed to homes, businesses, and industries.

Advantages and Dis-advantages of Wind Energy

ADVANTAGES OF WIND ENERGY

- Renewable and Sustainable
- Environmental Friendly
- Low Operating Costs
- Energy Independence

Disadvantages of Wind Energy:

→ Intermittent and Unpredictable

→ High Initial Costs

→ Impact on Wildlife

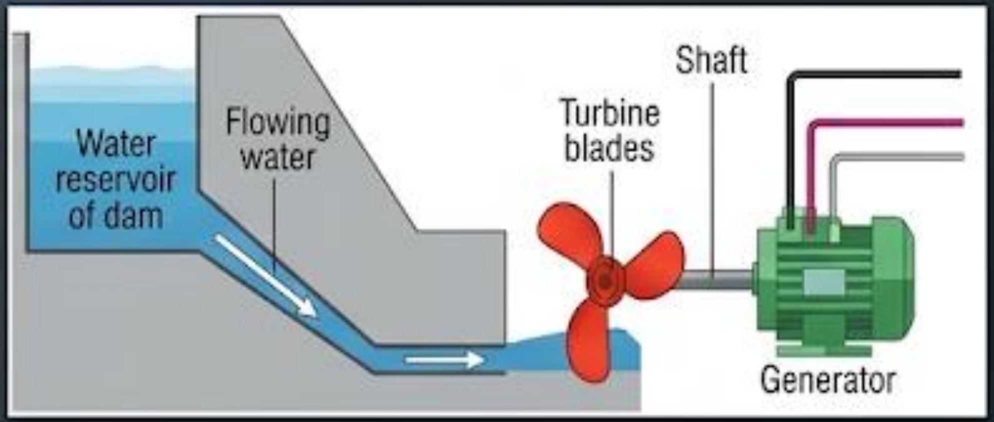
→ Noise

DISADVANTAGES OF WIND ENERGY

- Intermittent and Unpredictable
- High Initial Costs
- High Initial Costs
- Impact on Wildlife
- Noise

Non-conventional energy resources - Hydropower Energy

- Hydropower energy, or hydroelectric power, is a renewable energy source that generates electricity by using the energy of flowing water.
- Water is stored in the reservoir behind a dam.
- Controlled release of water flows through penstocks.
- The flowing water turns the turbine blades.
- The rotating turbine spins a connected shaft.
- The shaft drives the generator, producing electricity.



$$\text{HE} \xrightarrow[\text{Turbine}]{\text{Hydraulic}} \text{ME} \xrightarrow{\text{Generator}} \text{EE}$$

Advantages and Dis-advantages of Hydropower Energy

Advantages of Hydropower Energy:

→ Renewable and Sustainable ✓

→ Low Emissions ✓

→ Reliable and Stable ✓

→ Flexible ✓

Disadvantages of Hydropower Energy:

→ Environmental Impact ⚠

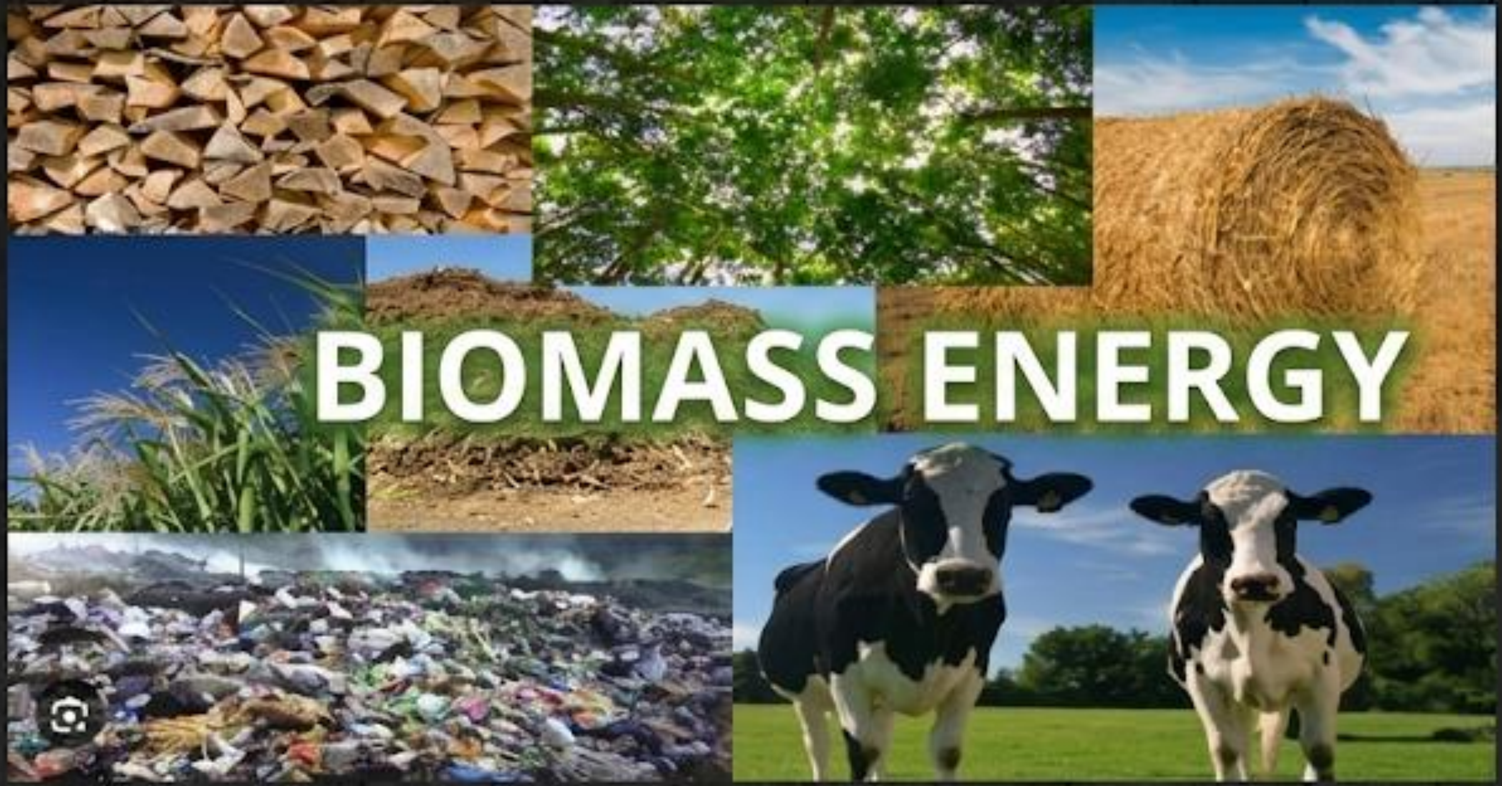
→ High Initial Costs ⚠

→ Dependence on Water Availability ⚠

→ Limited Suitable Sites ⚠

Non-conventional energy resources - Biomass energy

- Biomass energy is a type of renewable energy that is derived from organic materials.
- These materials, also known as biomass, include plant and animal waste, wood, crops, and other organic matter.
- The energy stored in these materials can be converted into electricity, heat, or biofuels.



Non-conventional energy resources - Biomass energy

SOURCES OF BIOMASS

Sources of Biomass:


Wood and Wood Waste


Agricultural Residues


Animal Manure


Industrial Waste

CONVERSION PROCESSES

Conversion Processes:

Combustion:
Burning biomass directly to produce heat or electricity.

Gasification:
Converting biomass into a gas (syngas) that can be burned for energy.

Anaerobic Digestion:
Breaking down organic matter in the absence of oxygen to produce biogas.

Fermentation:
Converting biomass into biofuels like ethanol through microbial processes.

Non-conventional energy resources-Biomass energy

Benefits of Biomass Energy:

- Renewable
- Carbon Neutral
- Waste Reduction

Applications:

- Electricity Generation
- Heating
- Transportation Fuels

Non-conventional energy resources-Geothermal energy

→ Geothermal energy is a form of renewable energy that is derived from the natural heat of the Earth.

→ This heat can be utilized for various applications, including electricity generation, direct heating, and industrial processes.

Non-conventional energy resources-Biomass energy

Sources of Geothermal Energy:

Hot Water Reservoirs: Naturally occurring pools of hot water located underground.

Geothermal Reservoirs: Deep underground sources of heat often found near tectonic plate boundaries.

Benefits of Geothermal Energy:

→ Renewable and Sustainable

→ Low emission

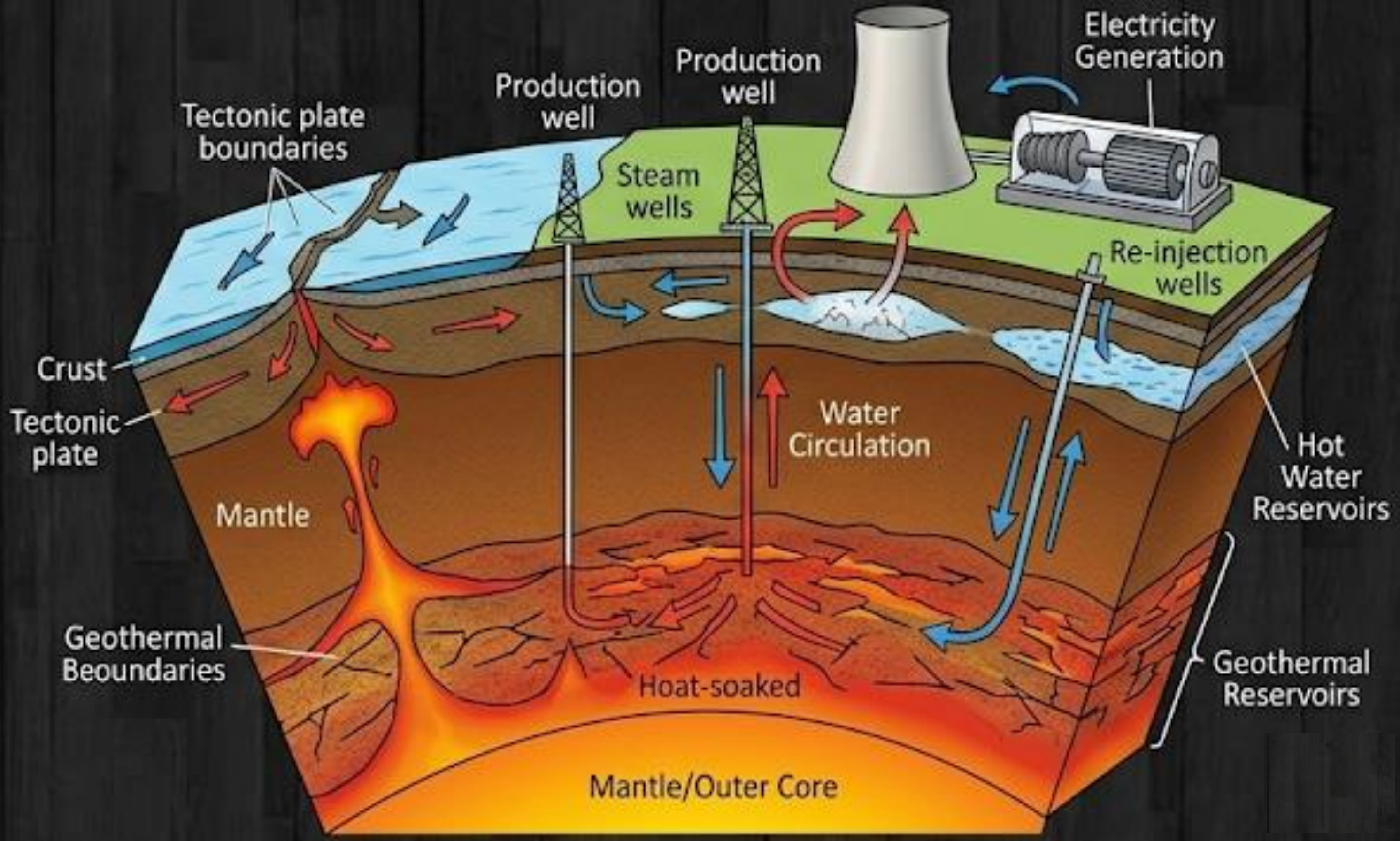
→ Base Load Energy

Challenges:

→ Location Specific

→ High Initial Costs

→ Environmental Concerns



Hydrogen is an alternative future source of energy

AKTU -2022-23

→ Hydrogen is a very light gas; its density is eight times less than that of natural gas.

→ There are no significant problems regarding storage, transportation, and dispensation.

→ Hydrogen is either formed through the steam reformation of natural gas or through electrolysis of water with renewable energies such as solar, wind and geothermal.

→ More than 90% of energy requirementts are met by fossil fuels, causing fuels which leads to emission of carbon dioxide and other toxic gases, which results in global warming. A hydrogen based transport system has the potential to play an important role in reducing greenhouse gas emissions.

→ To propel vehicles, hydrogen can be burnt directly in ICEs or used election engine (ICES) or can be used as fuel for producing electricity in fuel cells.

→ Fuel cell driven vehicles have great potential to be more efficient and eco-friendly than conventional fuel-driven vehicles,

→ The vehicles will only emit steam and will not emit any greenhouse gases.

Forest Resources

Forests, because of their ecological resources, play a vital role in human existence. Forests provide

indirect benefits by helping to sustain and control biophysical components of the ecosystem, such as climate and water.

→ Forests provide us with oxygen, shelter, jobs, water, food, nourishment and fuel.

→ Forests help to prevent erosion and conserve soil.

→ Forests, play an essential role in the biogeochemical control of the carbon and water cycles, the supply of wood and non-wood forest products.

→ Forests also play an important role in the global water cycle, moving water across the earth by releasing water vapor and capturing rainfall. They also filter out pollution and chemicals, improving the quality of water available for human use.

Deforestation

AKTU 2013-14, 2017-18, 2022-23

→ *Deforestation* is the large-scale removal of trees from forests (or other lands) for the facilitation of human activities.

→ It is a serious environmental concern since it can result in the loss of biodiversity, damage to natural habitats, disturbances in the water cycle, and soil erosion.

→ Deforestation is a major cause to climate change and global warming.

→ Deforestation has significant environmental, social, and economic impacts.

→ Forests cover approximately 31% of the total land surface of the Earth.

Causes or factors influencing Deforestation

AKTU 2013-14, 2017-18, 2023-24

1. Agricultural Expansion:   

→ The conversion of forests into agricultural land is a major cause of deforestation.

→ As the global demand for food increases, more land is cleared for farming, especially for crops like soy, palm oil, and cattle ranching.

2. Logging:  

→ The logging industry, both legal and illegal, is responsible for clearing large areas of forests.

→ Timber is in high demand for construction, furniture, and paper production.

3. Infrastructure Development:  

→ The construction of roads, highways, and urban areas often leads to deforestation as trees are cleared to make way for these projects.

→ This is particularly evident in developing countries undergoing rapid industrialization.

Causes or factors influencing Deforestation

4. Mining:

→ Extraction of minerals and other resources from forests contributes to deforestation.

→ This can result in habitat destruction and pollution, impacting both flora and fauna.

5. Forest Fires:

→ Natural and human-induced fires can cause extensive damage to forests, leading to deforestation.

→ In some cases, forests are intentionally set on fire for agricultural purposes, which can spiral out of control.

6. Climate Change:

→ Changing climatic conditions, including increased temperatures and altered precipitation patterns, can stress forests and make them more susceptible to diseases and pests.

→ This, in turn, may lead to deforestation as weakened trees are more vulnerable to logging or fires.

...Causes or factors influencing Deforestation

7. Population Pressure:

→ High population density in certain regions can contribute to increased demand for land and resources, leading to deforestation as a means of meeting these demands.

8. Government Policies and Regulations:

→ Weak or poorly enforced forest conservation policies can contribute to deforestation.

→ In some cases, policies that prioritize economic development over environmental conservation can lead to unsustainable land use practices.

Adverse effects of Deforestation

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Deforestation has numerous adverse effects on the environment, biodiversity, climate, and even human societies. Some of the major consequences are as the following:

1. Loss of Biodiversity (जैव विविधता की हानि।): 

→ Many species lose their habitats, leading to population declines and extinctions.

→ The balance of forest ecosystems is disturbed, affecting species interactions and ecosystem functions

2. Climate Change (जलवायु परिवर्तन): 

→ Forests play a crucial role in regulating the Earth's climate by absorbing and storing carbon dioxide (CO2).

→ When trees are cut down or burned, this stored carbon is released into the atmosphere, contributing to the greenhouse effect and global warming.

Adverse effects of Deforestation

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3. Disruption of Water Cycles (जल चक्र में व्यवधान):

→ Trees play a vital role in maintaining local and regional water cycles. Deforestation can disrupt these cycles, leading to altered precipitation patterns, reduced water quality, and increased risk of floods and droughts.

4. Soil Erosion (मृदा अपरदन):

→ Tree roots help bind the soil together, preventing erosion. When trees are removed, especially on slopes, the soil becomes more susceptible to erosion.

→ This can result in landslides, decreased soil fertility, and sedimentation of water bodies.

Adverse effects of Deforestation

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5. Increased Greenhouse Gas Emissions:

→ Beyond the immediate release of carbon stored in trees, deforestation can lead to ongoing emissions as the cleared land is often used for agriculture or other purposes that involve the use of fossil fuels and other emission sources.

6. Impact on Indigenous Peoples (आदिवासी लोगों पर प्रभाव):

→ Indigenous communities often rely on forests for their livelihoods, cultural practices, and sustenance.

→ Deforestation can threaten their way of life, leading to displacement, loss of resources, and challenges to their cultural heritage.

...Adverse effects of Deforestation

AKTU 2013-14, 2017-18, 2023-24

7. Decreased Rainfall:

→ Forests contribute to the formation of clouds and precipitation.

→ The removal of trees can disrupt local weather patterns, potentially leading to decreased rainfall in the affected areas.

8. Air and Water Pollution:

→ Deforestation can contribute to increased pollution as the absence of trees allows pollutants to reach water bodies more easily.

→ Without the natural filtration provided by forests, water quality may deteriorate..

AKTU : B.Tech (III-Sem)

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Branches : ME & ME Allied

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Thank You

For your commitment to a sustainable future